

Lab - Introduction to Amazon DynamoDB

Lab Overview

Amazon DynamoDB is a fast and flexible NoSQL database service for all applications that need consistent, single-digit millisecond latency at any scale. It is a **fully managed database** and supports both document and key-value data models. Its flexible data model and reliable performance make it a great fit for mobile, web, gaming, ad-tech, IoT, and many other applications.

In this lab, you create a table in Amazon DynamoDB to store information about a music library. You then query the music library and, finally delete the DynamoDB table.

Objectives

After completing this lab, you will be able to:

- Create an Amazon DynamoDB table.
- Enter data into an Amazon DynamoDB table.
- Query an Amazon DynamoDB table.
- Delete an Amazon DynamoDB table.

Start lab

1. To launch the lab, at the top of the page, choose **Start Lab**.

Caution: You must wait for the provisioned AWS services to be ready before you can continue.

2. To open the lab, choose **Open Console**.

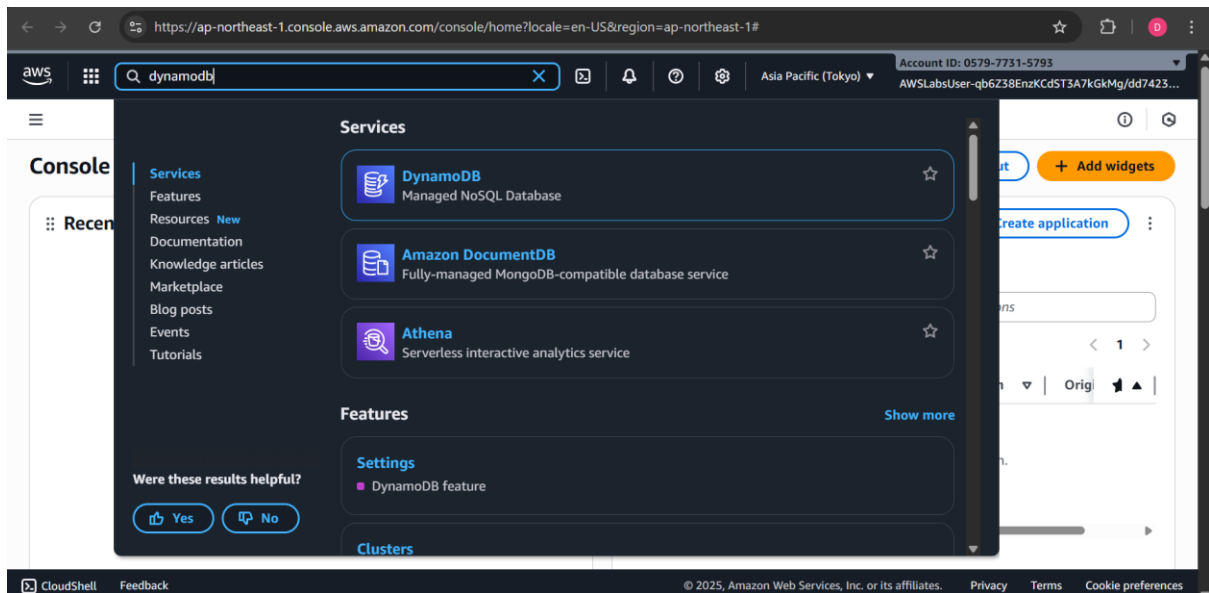
You are automatically signed in to the AWS Management Console in a new web browser tab.

Warning: Do not change the **Region** unless instructed.

Task 1: Create a New Table

In this task, you create a new table in DynamoDB named *Music*. Each table requires a Primary Key that is used to partition data across DynamoDB servers. A table can also have a Sort Key. The combination of Primary Key and Sort Key uniquely identifies each item in a DynamoDB table.

3. At the top of the AWS Management Console, in the search bar, search for and choose **DynamoDB**



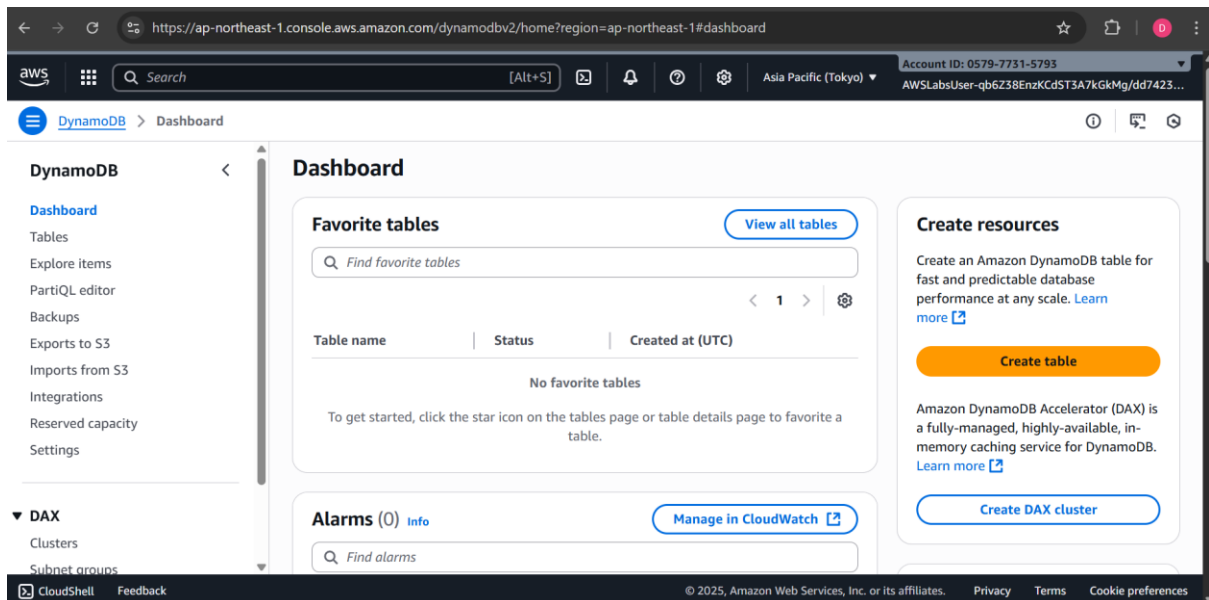
4. On the **Amazon DynamoDB** getting started page, choose **Create table**.

On the **Create table** page, configure the following options:

In the **Table details** section:

- For **Table name**, enter Music
- For **Partition key**, enter Artist and choose **String** from the dropdown menu.
- For **Sort key - optional**, enter Song and choose **String** from the dropdown menu.

Your table uses default settings for indexes and provisioned capacity.



← → ↻ https://ap-northeast-1.console.aws.amazon.com/dynamodbv2/home?region=ap-northeast-1#create-table

aws Search [Alt+S] Asia Pacific (Tokyo) Account ID: 0579-7731-5793 AWSLabUser-qb6Z38EnzKCdST3A7kGkMg/dd7423...

DynamoDB > Tables > Create table

Table details [Info](#)

DynamoDB is a schemaless database that requires only a table name and a primary key when you create the table.

Table name
This will be used to identify your table.

Music

Between 3 and 255 characters, containing only letters, numbers, underscores (_), hyphens (-), and periods (.).

Partition key
The partition key is part of the table's primary key. It is a hash value that is used to retrieve items from your table and allocate data across hosts for scalability and availability.

Artist String

1 to 255 characters and case sensitive.

Sort key - optional
You can use a sort key as the second part of a table's primary key. The sort key allows you to sort or search among all items sharing the same partition key.

Song String

1 to 255 characters and case sensitive.

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5. Choose **Create table**.

Expected output: Initial message.

A Creating the Music table. It will be available for use shortly. message is displayed on top of the screen.

6. Wait for your table to be created and its status to be Active.

Congratulations! You have successfully created a new table named *Music* in DynamoDB.

← → ↻ https://ap-northeast-1.console.aws.amazon.com/dynamodbv2/home?region=ap-northeast-1#tables

aws Search [Alt+S] Asia Pacific (Tokyo) Account ID: 0579-7731-5793 AWSLabUser-qb6Z38EnzKCdST3A7kGkMg/dd7423...

DynamoDB > Tables

DynamoDB

- Dashboard
- Tables**
- Explore items
- PartiQL editor
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▼ **DAX**

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Tables (2) [Info](#)

Find tables Any tag key Any tag value

	Name	Status	Partition key	Sort key	Indexes	Replication Regions	Deletion protection
☐	LabEnforcer_Table	Info					Unavailable
☐	Music	Active	Artist (S)	Song (S)	0	0	Off

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ask 2: Add Data

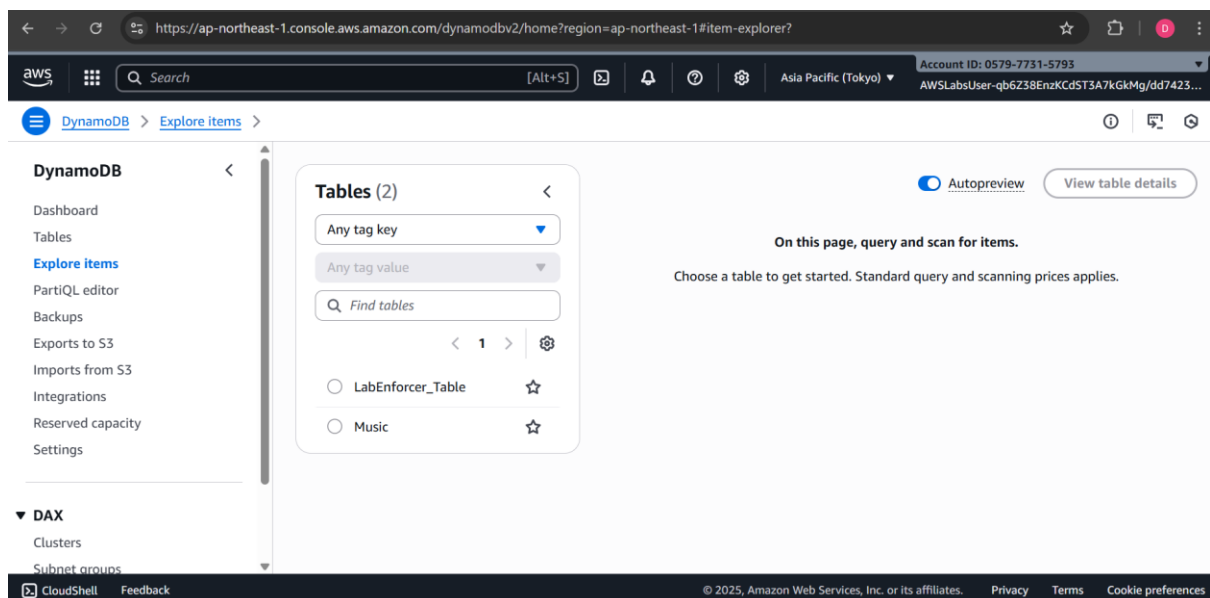
In this task, you add data to the *Music* table. A **table** is a collection of data on a particular topic.

Each table contains multiple **items**. An item is a group of attributes that is uniquely identifiable among all of the other items. Items in DynamoDB are similar in many ways to rows in other database systems. In DynamoDB, there is no limit to the number of items you can store in a table.

Each item is composed of one or more **attributes**. An attribute is a fundamental data element, something that does not need to be broken down any further. For example, an item in a *Music* table contains attributes such as Song and Artist. Attributes in DynamoDB are similar columns in other database systems, but each item (row) can have different attributes (columns).

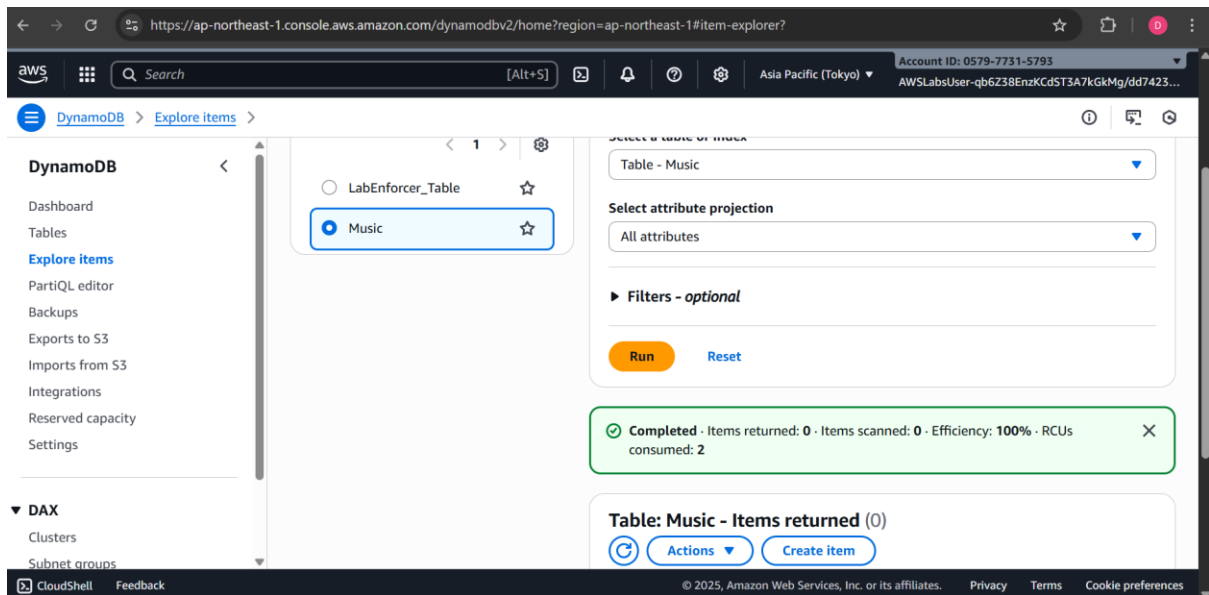
When you write an item to a DynamoDB table, only the Primary Key and Sort Key (if used) are required. Other than these fields, the table does not require a schema. This means that you can add attributes to one item that may be different to the attributes on other items.

7. In the left navigation pane, choose **Explore items**.



8. Select **Music** table.
9. Choose **Create item**.

On the **Create item** page, configure the following options:



10. For **Artist** Value, enter Pink Floyd

11. For **Song** Value, enter Money

These are the only required attributes, but you also add additional attributes.

12. To create an additional attribute, choose **Add new attribute** .

13. In the drop-down list, select **String**.

A new *string* attribute row gets added.

14. For the new attribute, enter:

- For **Attribute name**, enter Album
- For **Value**, enter The Dark Side of the Moon

15. Create another new attribute by choosing **Add new attribute** .

16. In the drop-down list select **Number**.

A new *number* attribute gets added.

17. For the new attribute, enter:

- For **Attribute name**, enter Year
- For **Value**, enter 1973

18. Choose **Create item** to store the new item with its four attributes.

Create item

You can add, remove, or edit the attributes of an item. You can nest attributes inside other attributes up to 32 levels deep. [Learn more](#)

Attributes

Attribute name	Value	Type	
Artist - Partition key	Pink Floyd	String	
Song - Sort key	Money	String	
Album	The Dark Side of the Moon	String	Remove
Year	1973	Number	Remove

[Add new attribute](#) [Cancel](#) [Create item](#)

Expected output:

A The item has been saved successfully. message is displayed on top of the screen.

The item now appears in the console.

The item has been saved successfully.

DynamoDB

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▼ DAX

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Tables (2)

- Any tag key
- Any tag value
-
- 1
- LabEnforcer_Table
- Music**

Music

☒ Autopreview [View table details](#)

▼ Scan or query items

☒ Scan ☐ Query

Select a table or index

Table - Music

Select attribute projection

All attributes

► Filters - optional

19. Create a second item, using these attributes:

Attribute Name	Attribute Type	Attribute Value
Artist	String	John Lennon
Song	String	Imagine
Album	String	Imagine
Year	Number	1971

Attribute Name	Attribute Type	Attribute Value
Genre	String	Soft rock

The screenshot shows the AWS DynamoDB console interface for creating an item in a table named 'Music'. The 'Attributes' section lists the following attributes:

Attribute name	Value	Type	Action
Artist - Partition key	John Lennon	String	
Song - Sort key	Imagine	String	
Album	Imagine	String	Remove
Year	1971	Number	Remove
Genre	Soft rock	String	Remove

Note that this item has an additional attribute called *genre*. This is an example of each item being capable of having different attributes without having to pre-define a table schema.

20. Create a third item, using these attributes:

Attribute Name	Attribute Type	Attribute Value
Artist	String	Psy
Song	String	Gangnam Style
Album	String	Psy 6 (Six Rules), Part 1
Year	Number	2011
LengthSeconds	Number	219

Attributes

Attribute name	Value	Type	
Artist - Partition key	Psy	String	
Song - Sort key	Gangnam Style	String	
Album	Psy 6 (Six Rules), Part 1	String	Remove
Year	2011	Number	Remove
LengthSeconds	219	Number	Remove

Cancel Create item

Once again, this item has a new *LengthSeconds* attribute identifying the length of the song. This demonstrates the flexibility of a NoSQL database.

There are also faster ways to load data into DynamoDB, such as using AWS Data Pipeline, programmatically loading data or using one of the free tools available on the Internet.

Congratulations! You have successfully added data to the *Music* table.

Task 3: Modify an Existing Item

In this task, you modify an existing item.

21. From the list of items, select **Psy**.
22. Choose **Actions**.
23. Choose **Edit item**.

Completed - Items returned: 0 - Items scanned: 0 - Efficiency: 100% - RCUs consumed: 2

Table: Music - Items returned (1/3)

Actions

- Edit item
- Duplicate item
- Delete items
- Download selected items to CSV
- Download results to CSV

Artist	Song	Album	Year	LengthSeconds
Psy	Gangnam Style	Psy 6 (Six Rules), Part 1	2011	219
John Lennon	Imagine	Imagine	1971	180
Pink Floyd	Money	The Dark Side of the Moon	1973	216

24. Change the **Year** from **2011** to **2012**.

Attributes

Attribute name	Value	Type	
Artist - Partition key	Psy	String	
Song - Sort key	Gangnam Style	String	
Album	Psy 6 (Six Rules), Part 1	String	Remove
LengthSeconds	219	Number	Remove
Year	2012	Number	Remove

Cancel Save Save and close

25. Choose **Save changes**.

Expected output:

A The item has been saved successfully. message is displayed on top of the screen.

The item is now updated.

Congratulations! You have successfully updated an item within the *Music* table.

The item has been saved successfully.

DynamoDB

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Tables (2)

- Any tag key
- Any tag value
- Find tables
- 1
- LabEnforcer_Table
- Music

Music

Autopreview View table details

▼ Scan or query items

Scan Query

Select a table or index

Table - Music

Select attribute projection

All attributes

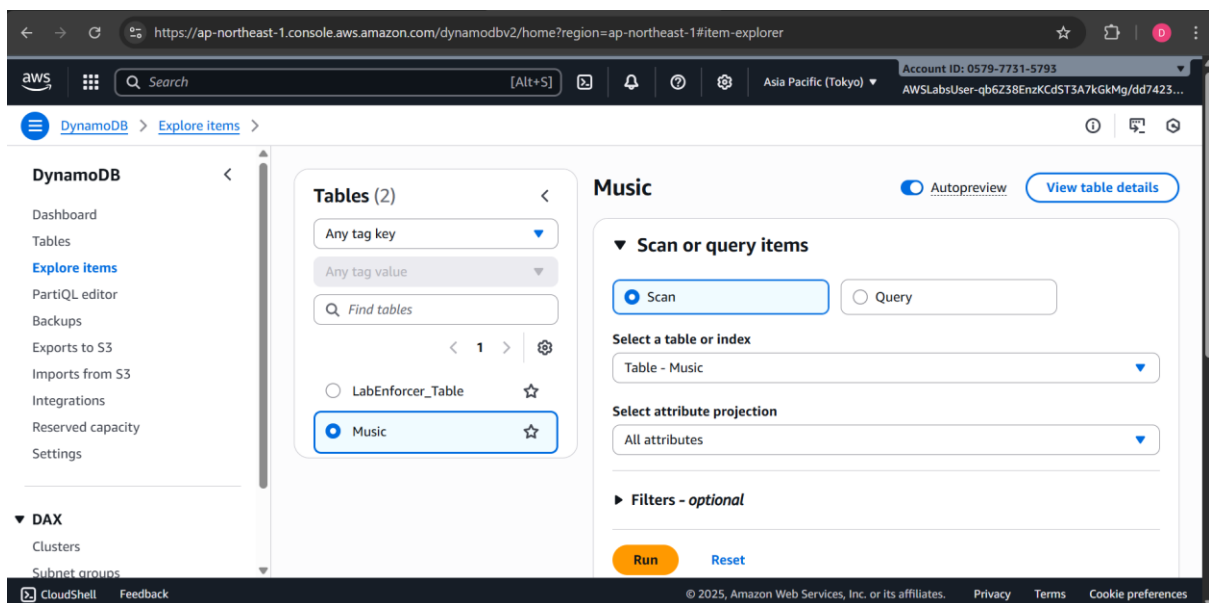
► Filters - optional

Task 4: Query the Table

In this task, you query the *Music* table. There are two ways to query a DynamoDB table: *Query* and *Scan*.

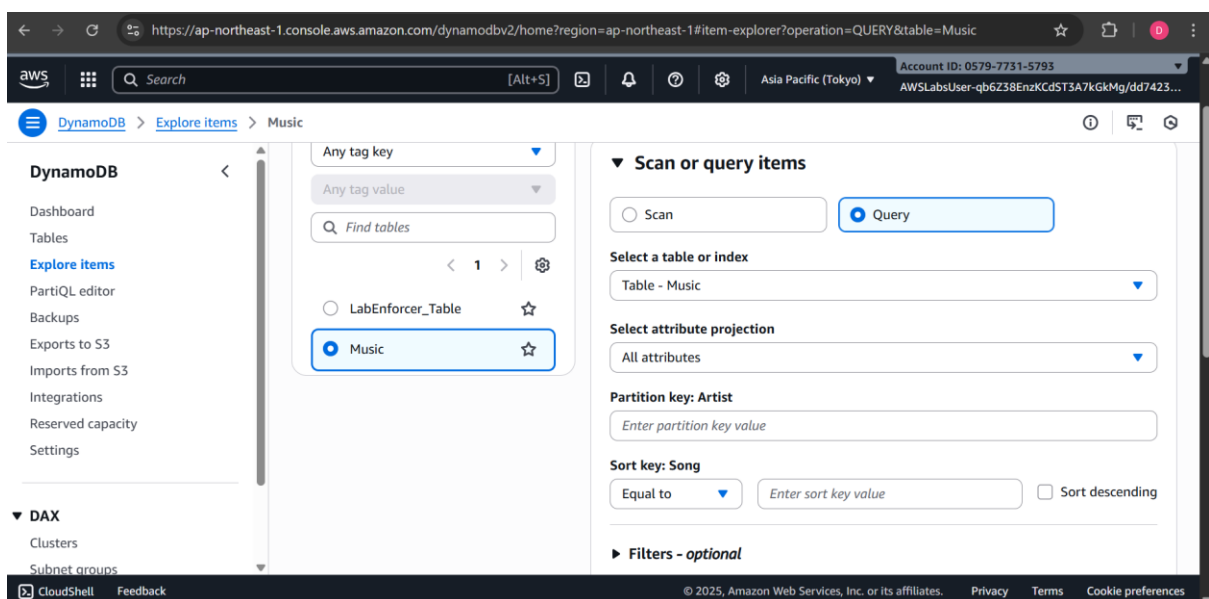
A **query** operation finds items based on Primary Key and optionally Sort Key. It is fully indexed, so it runs very fast.

26. In the left navigation pane, choose **Explore items**.
27. Select **Music** table.
28. Expand **Scan or query items** to query or scan items.



29. Select **Query**.

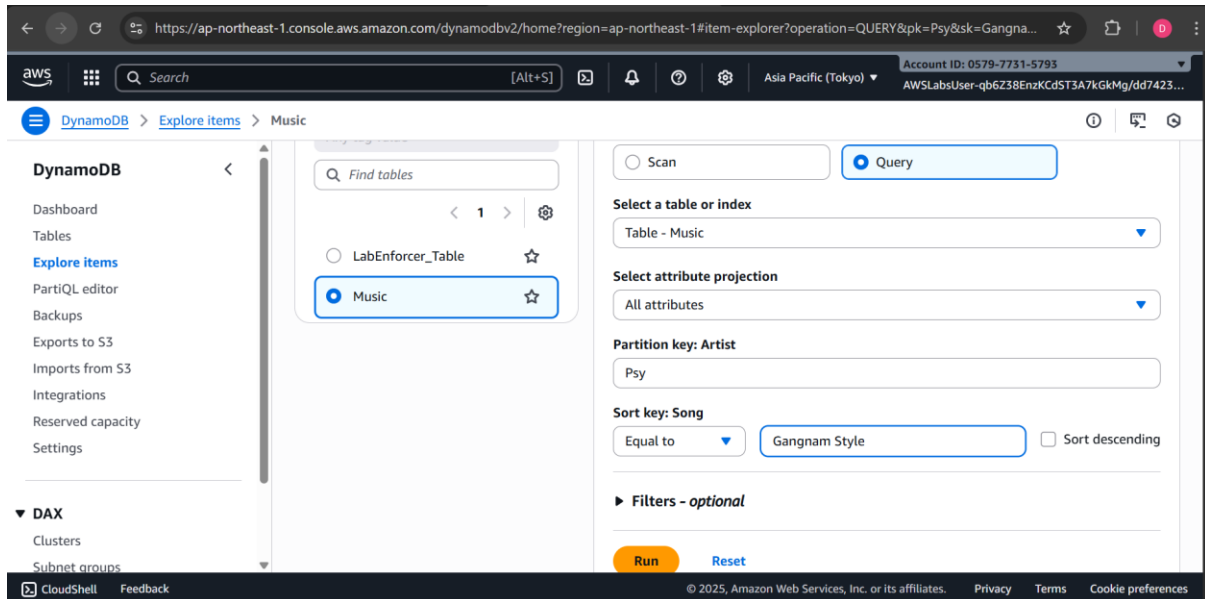
Fields for the Partition Key (which is the same as Primary Key) and Sort Key are now displayed.



30. Configure the following for Partition and Sort keys

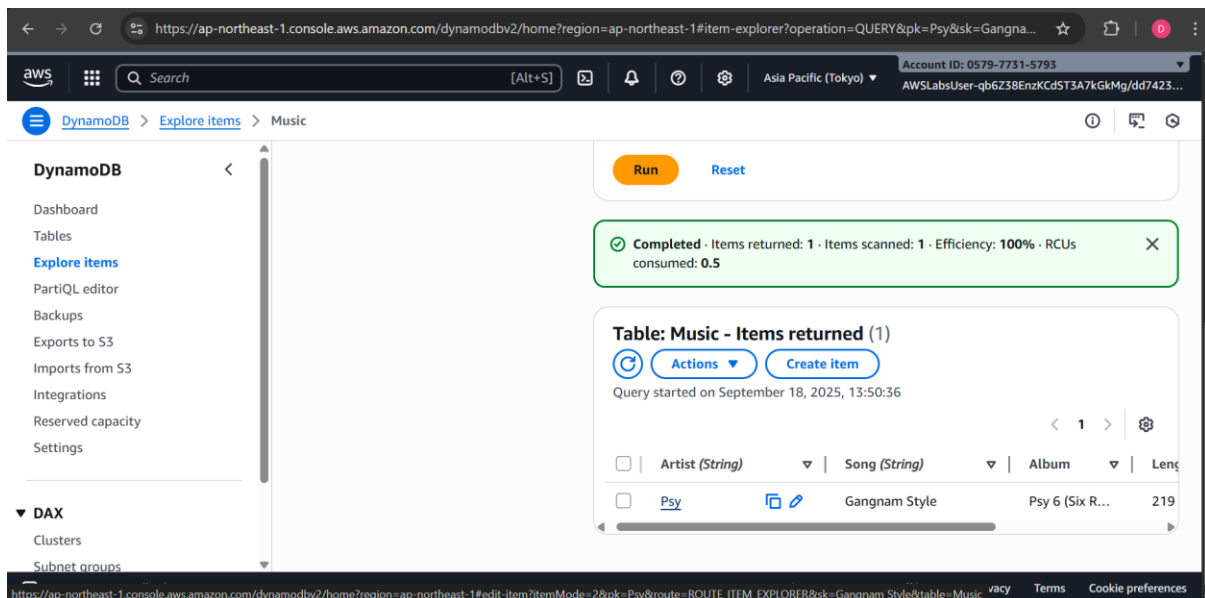
- For **Artist (Partition key)**, enter Psy
- For **Song (Sort key)**, select **Equal to** from the dropdown menu and enter Gangnam Style

31. Choose **Run**.



Expected output:

The song quickly appears in the list. A *query* is the most efficient way to retrieve data from a DynamoDB table.



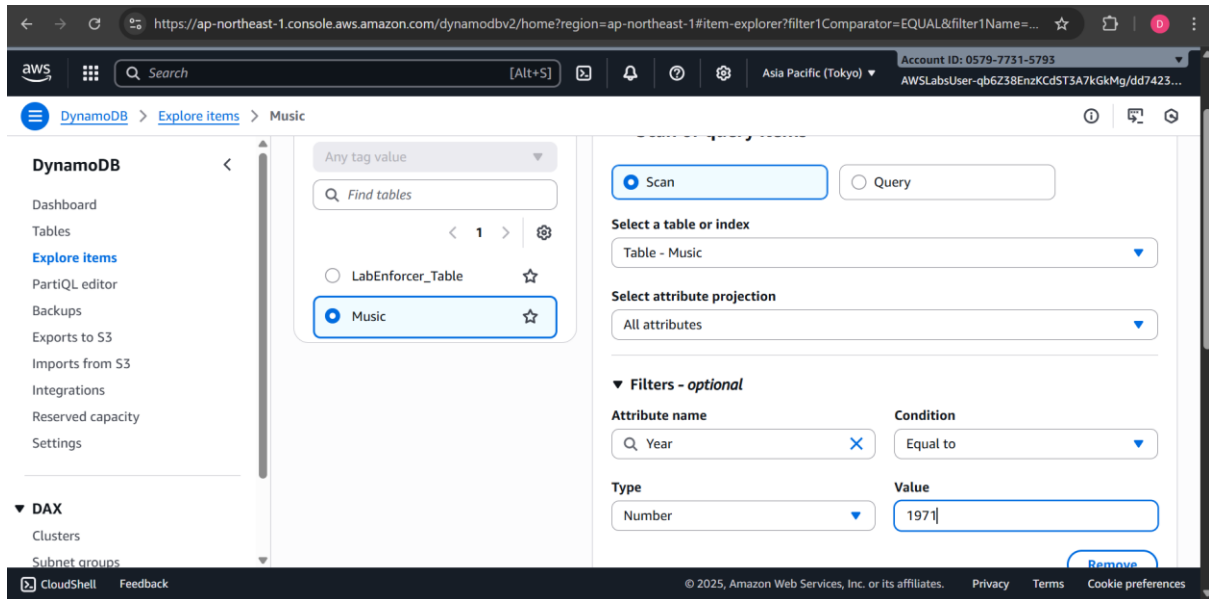
Alternatively, you can *scan* for an item. This involves looking through *every item in a table*, so it is less efficient and can take significant time for larger tables.

32. Select **Scan**.

33. Expand the **Filters** section, and configure the following:

- For **Attribute name**, enter Year

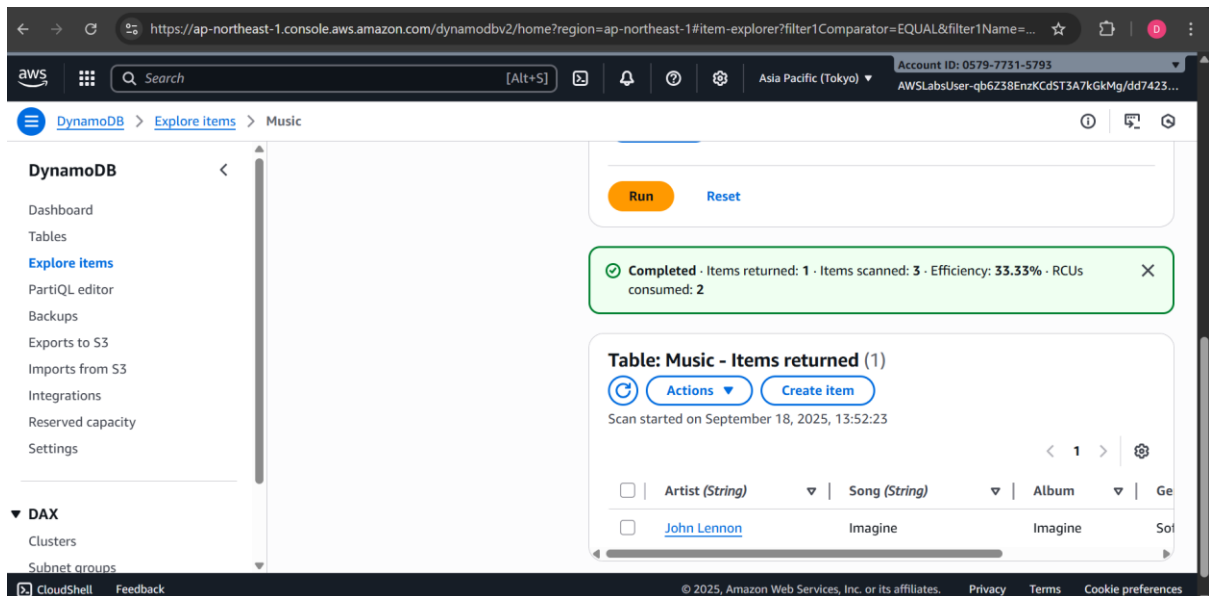
- For **Type**, select **Number** from the dropdown menu.
- For **Condition**, select **Equal to** from the dropdown menu.
- For **Value**, enter 1971



34. Choose **Run**.

Expected output:

Only the song released in 1971 is displayed.



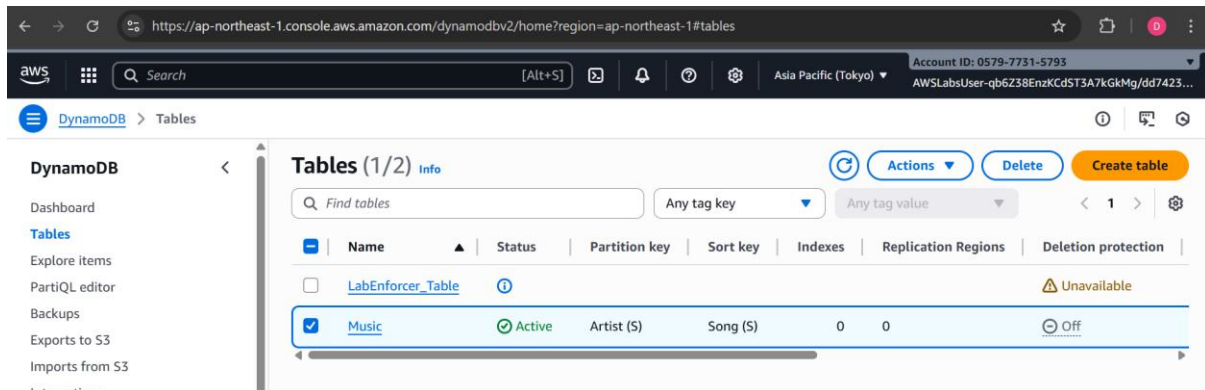
Congratulations! You have successfully used the *query* and *scan* operations to query the *Music* table.

Task 5: Delete the Table

In this task, you delete the *Music* table, which also deletes all the data in the table.

35. In the left navigation pane, choose **Tables**.

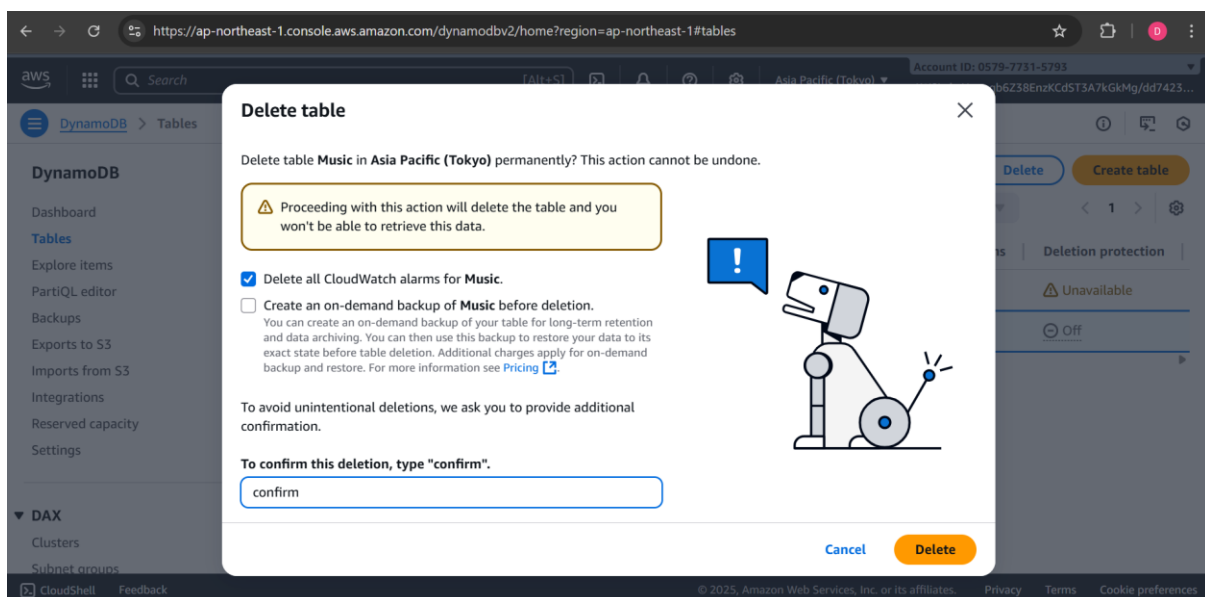
36. Select **Music** table.



37. Choose **Delete**.

38. On the **Delete table** pop-up window, locate the **Enter “confirm” to agree.** textbox and enter confirm

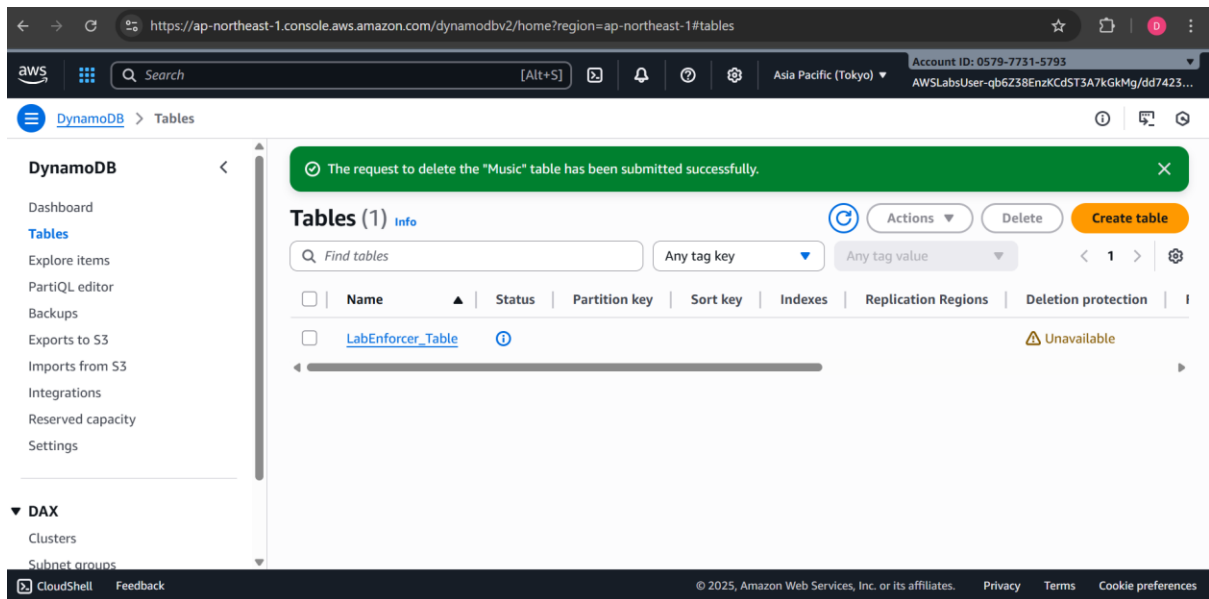
39. Choose **Delete**.



Expected output: Initial message.

A The request to delete the “Music” table has been submitted successfully. message is displayed on top of the screen.

40. Use the refresh option to confirm the table deletion.



Congratulations! You have successfully deleted the *Music* table.

Conclusion

Congratulations! You have now successfully learned how to:

- Create an Amazon DynamoDB table.
- Enter data into an Amazon DynamoDB table.
- Query an Amazon DynamoDB table.
- Delete an Amazon DynamoDB table.

End lab

Follow these steps to close the console and end your lab.

41. Return to the **AWS Management Console**.
42. At the upper-right corner of the page, choose **AWSLabsUser**, and then choose **Sign out**.
43. Choose **End Lab** and then confirm that you want to end your lab.

Additional Resources

- [Amazon DynamoDB Documentation](#)